

Training Day 4 Report

Date: 20 June 2025

Module Name: Network Fundamentals & OSI Model

Mode: Online

Main Focus of the Module

The fourth day of training introduced key concepts in **computer networks**, focusing on **IP addressing**, **subnetting**, and the **OSI model**. We performed **hands-on subnet calculations** for both **classful** and **classless** addressing schemes, and understood the role of each OSI layer in network communication.

Topics Covered

Introduction to Computer Networks

- A **computer network** is a group of interconnected devices that share data.
 - Devices communicate over physical or wireless mediums using standardized protocols.
 - Basic network types: LAN, WAN, MAN, PAN
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☐ IP Addressing – Classful Addressing

Class	Range	Default Subnet Mask	Network Bits	Host Bits	No. of Hosts	Usage
B	128.0.0.0 – 191.255.255.255	255.255.0.0	16	16	$2^{16} - 2 = 65,534$	Medium networks
C	192.0.0.0 – 223.255.255.255	255.255.255.0	24	8	$2^8 - 2 = 254$	Small networks

Class B Example

IP Address: 172.16.0.0

- **Subnet Mask:** 255.255.0.0 (/16)
- **Network Bits:** 16
- **Host Bits:** 16
- **First Host IP:** 172.16.0.1

- **Last Host IP:** 172.16.255.254
 - **Broadcast IP:** 172.16.255.255
 - **Total Hosts:** 65,534
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Class C Example

IP Address: 192.168.1.0

- **Subnet Mask:** 255.255.255.0 (/24)
 - **Network Bits:** 24
 - **Host Bits:** 8
 - **First Host IP:** 192.168.1.1
 - **Last Host IP:** 192.168.1.254
 - **Broadcast IP:** 192.168.1.255
 - **Total Hosts:** 254
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Classless Inter-Domain Routing (CIDR)

- Allows variable-length subnet masking (VLSM)
- No strict boundary of A/B/C classes
- Represented as /x where x is number of network bits

Example:

IP: 192.168.10.0/26

- **Subnet Mask:** 255.255.255.192
 - **Network Bits:** 26
 - **Host Bits:** 6
 - **Total Hosts:** $2^6 - 2 = 62$
 - **Valid Range:** 192.168.10.1 to 192.168.10.62
 - **Broadcast IP:** 192.168.10.63
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☐ Hands-On Practice

We practiced identifying the following from given IPs:

- Subnet mask
- Network ID
- Broadcast address
- Valid host range
- Total number of networks and hosts in subnet
- Calculated IP ranges for multiple subnets

Introduction to the OSI Model

The **OSI (Open Systems Interconnection)** model has 7 layers that define network communication:

Layer	Name	Function
7	Application	User interface (HTTP, FTP)
6	Presentation	Data encoding/encryption
5	Session	Session management
4	Transport	Reliable delivery (TCP/UDP)
3	Network	IP addressing and routing
2	Data Link	MAC addressing, switching
1	Physical	Hardware and transmission media

Key Takeaways

- Understood how IP addressing and subnetting determine valid hosts and networks
- Learned to perform subnetting manually using classful and classless methods
- Grasped the purpose of each layer of the OSI model and its real-world protocols
- Recognized how addressing and subnetting are fundamental for ethical hacking and penetration testing